

A Causal Analysis of Major Lethal Aviation Accidents and the Effect of Present Contributing Factors

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Abstract

This report investigates the causal factors that influence major lethal aviation accidents using a causal inference approach. Traditional aviation safety often relies on descriptive statistics to correlate results, but it fails at finding the dependencies between the contributing factors. To mitigate this, causal structure techniques are used within this report to better understand lethality influences.

The dataset used consists of 5,268 aviation accidents, including structured variables and text summaries, from 1908 to 2009. A keyword classification algorithm was developed to find and categorize contributing factors. Factors may include but are not limited to weather, mechanical failure, pilot error, and fire. These categorized factors were then used to determine a model.

A FCI (Fast Causal Inference) algorithm combined with Chi-Squared conditional independence test was used to create PAGs or partially ancestral graphs. The PAGs identified causal relationships using latent confounders and constraints to further refine the models. Finally, linear regression with backdoor adjustments quantified each factor's impact on lethality.

Results indicate several factors positively impacted accident severity, such as military status, Controlled Flight into Terrain (CFIT), and weather, while fuel related incidents and mechanical failure were associated with lower lethality. Overall, findings highlight causal modeling is more insightful than its correlation counterparts as it uses causal drivers rather than coincidental associations.

This work demonstrates that causal discovery is an effective method for analyzing aviation lethality from this dataset. By identifying key drivers of highly lethal accidents, this report provides valuable insight that can positively impact the safety and future of the aviation industry.

Problem Overview

Ever since the first recorded airplane fatality on September 17, 1908, aviation accidents have driven thousands of deaths and, in response, some of the most consequential safety innovations in modern transportation. Identifying the causal structure underlying these accidents offers a principled basis for distinguishing which factors drive accident severity and which are incidental [1].

The dataset available for this analysis comprises 5,268 fatal aviation accidents occurring between 1908 and 2009. Because the dataset contains only accidents in which at least one fatality occurred, the research question cannot address the causes of accidents. Instead, this analysis investigates which contributing factors present in the accident description portion of the dataset influence the severity of fatal accidents. Severity is approximated as a normalized fatality rate to allow accidents to allow more effective direct comparison. Accidents with a normalized fatality rate exceeding 0.5 are classified as high-lethality; those at or below 0.5 are classified as low-lethality. Single-occupant, single-fatality records are excluded, as their fatality rate is trivially 1.0 and their inclusion in the dataset is a selection artifact rather than an informative observation.

Contributing factors were extracted from unstructured textual accident summaries using a keyword classification algorithm. Each accident was assigned to one of ten user defined categories: sabotage, shot down, fire, fuel, mechanical failure, weather, collision, controlled flight into terrain (CFIT), pilot error, and unknown. This was based on a ranked keyword-frequency approach. An eleventh category, undetermined, was initially included but later excluded from the causal model on the grounds that it reflects investigative limitations rather than a substantive contributing factor.

To recover the causal structure among these factors and their relationship to accident severity, the fast causal inference (FCI) algorithm, as implemented in the causal-learn library [2], was used to estimate a (partial ancestral graph) PAG from the available observational data. Contextual knowledge was encoded as background constraints. With the resulting causal graph, the adjustment sets necessary for estimating causal effects were identified using the DoWhy Library [3], and DoWhy’s built in linear regression function was applied to quantify the effect of each contributing factor on the probability of high-severity outcomes, conditional on satisfaction of the backdoor criterion.

Background and Literature

Understanding the Lethality of crashes is vital in improving Aviation safety. While identifying the cause of a crash can help prevent future incidents, identifying the causes of fatalities during said crash can help people survive if the crash does occur. A commentary on Crash Risk in General Aviation [4] helps explain the significance of identifying the causes of fatalities during crashes. While fatality rates per crash in commercial aviation went down significantly between 1987 and 2007, the fatality rate per crash in general aviation stayed at 19% over the same 20 years. This meant that although the chances of a crash were lower in general aviation, the survival chances of a person during a general aviation crash in 1987 and 2007 were roughly the same.

Several factors were identified as major factors in how fatal a crash was. The largest contributor was the presence of fire. Fires occurred in only 13% of all general aviation incidents. Without further analysis, this might indicate that fire is not a major contributor. However, fires account for 40% of the total deaths in general aviation. Only 15% of general aviation incidents that occurred without fires resulted in fatalities, meanwhile 69% of general aviation incidents that included a fire resulted in fatalities. These statistics highlight an area where engineering a solution would save a significant number of lives, reducing the risk of fires through engineering improved piping, fire suppression methods, and emergency escape methods could either lower the rates of fires, or reduce the fatality rates when a fire is present.

Statistics can not only expose engineering shortcomings, but human factors. Not wearing safety restraints was also identified as a major cause in the commentary. Passengers who were not wearing shoulder safety harnesses were four times as likely to die during a crash. A gap between airframe complexity and pilot experience with the airframe was also identified. Physician pilots were identified as crashing at a higher rate than other pilots. This was due to the complexity of the airframes often used in the profession for rapid transport, and the available hours for the pilots to learn to operate the aircraft safely and familiarize themselves with aircraft systems.

While the commentary is from 2007, its comments on general aviation are still relevant. According to the NTSB crash statistics from 2024, general aviation fatality rates are still higher than commercial at 16%. By identifying the causes of mass fatality events, it may be possible to work to reduce that fatality rate even further.

Data Source

The Airplane Crashes and Fatalities dataset since 1908 was retrieved from Kaggle. This dataset captures 5,268 aviation accidents and fatalities incurred globally between 1908 and 2009, with one row per crash and columns describing when, where, and how the crash occurred, along with how many people were aboard, killed, and killed on the ground. As such, over a century of crash data can be utilized for analysis, spanning the earliest days of powered flight through the modern commercial aviation era.

For each recorded crash record, important features include date, time, location, operator, number of passengers aboard, fatalities aboard, ground fatalities, and a crash summary. The date and time fields record when the accident occurred, though time is not always present for earlier records. The location field captures where the crash took place, ranging from specific towns and cities to remote wilderness and open ocean, and requires geographic classification before it can be used analytically. The operator field identifies the airline, military branch, or private party responsible for the flight, and was used to derive the military status variable central to this analysis. The aboard and fatalities columns provide the raw figures used to compute the normalized fatality rate that serves as the outcome variable. Ground fatalities captures deaths among people not aboard the aircraft, such as those on the ground struck by wreckage, and are included in the normalized fatality calculation. The summary field contains a free-text description of each accident, written from an investigative perspective, and serves as the primary source for contributing factor classification throughout this project.

It is important to note several data quality considerations. Not all records contain a summary, and some summaries are too sparse to yield meaningful classification, such as entries reading only "crashed en route." Earlier records from the first few decades of aviation tend to have less detail overall, reflecting the less formalized accident investigation practices of the era. Record completeness improves markedly in the post-World War II period, coinciding with the establishment of formal aviation safety regulatory bodies such as the NTSB in the United States and equivalent agencies internationally..

Additionally, because the dataset includes only accidents in which at least one fatality occurred, it introduces an inherent selection bias: non-fatal accidents are entirely absent, which constrains the analysis to questions of accident severity rather than accident likelihood. This means the findings of this report speak to what makes a fatal accident more or less lethal, not to what makes an accident more or less likely to occur in the first place. This distinction is critical to the interpretation of all results presented in this report and informed several of the modeling decisions described in the Methodology section, including the exclusion of single-occupant, not to what makes an accident more or less likely to occur in the first place.

Methodology

The mythology of the project has three phases: Labeling, Causal Discovery, and Regression. Given the dataset provided the first phase is to label and format the data, so it is suitable for causal discovery and regression. Labeling was done utilizing keyword matching and then verified against a human annotated key using a confusion matrix to quantify the results.

Data Labeling

The dataset provides details on each aircraft accident with structured data columns, and the unstructured data column labeled summary. The structured data column provides simple, and ready-to-use data that can be used for analysis. Unstructured data requires processing, most often classification into different categories before analysis can be performed. The data that requires processing is labeled in Table 1.

Dataset Element Title	Data Description	Requires Processing?
Date	Date of the Accident	No
Time	Time of the Accident	No
Location	Location of the Accident	Yes
Operator	Operator of Aircraft	Yes
Flight #	Flight Number	No
Route	Departure and Arrival Airports	No
Type	Model of Aircraft	Yes
Registration	Registration Number	No
CN/IN	Aircraft Serial Number	No
Aboard	Souls Aboard Aircraft	No
Fatalities	Fatalities from Aircraft	No
Ground	Ground Fatalities	No
Summary	Extra Accident Information – Cause	Yes

Table 1: Which datapoints require processing

Contributing Factor Classification from Summary

The datapoint that requires the most processing is the summary datapoint. The summary datapoint contains the contributing factors that were involved in any specific accident. Utilizing keywords found in the summary datapoints, Python RegEx package was utilized to accomplish the character manipulation [5]. To construct the keyword list, all the significant words were ranked on how often they appear, and then the top 100 were sorted into likely associated contributing factors using Claude Sonnet-4.6 to generate possible keys from the list of top 100 words with manual human pruning and review to finalize. This process was then repeated four times, excluding words that were already sorted into a classification rule, until further additions did not yield more than a 10% reduction in “unknown” classifications. There is an inherent limit to this approach, given that not all accidents have a summary datapoint or sufficient details, i.e., “crashed en route”. The keywords derived can be seen in Table 2.

Contributing Factor	Associated Keywords
Sabotage	hijacked, hijackers, hijackers, hijackers, hijackers, hijacker, explosive, bomb, dynamite, sabotage, suicide, stolen, grenade
Shot Down	shot down, shot by, enemy fire, anti-aircraft, shot by fighter, fighter fire, gunfire, bullet
Fire	fire, flames, burned, burnt, exploded, explosion
Fuel	fuel exhaustion, fuel starvation, ran out of fuel, low fuel, out of fuel
Mechanical	engine failure, engine failed, mechanical failure, engine, engines, failure, failed, power loss, power failure, broken, malfunctioning, control problems, landing gear, gear, wing, fatigue, rudder, elevator
Weather	weather, fog, rain, visibility, adverse, conditions, vfr, thunderstorm, ice, wind, storm, snow, turbulence, windshear, crosswind, rainstorm, snowstorm, clouds
Collision	mid-air collision, midair collision, collided with another aircraft, collided with another airplane, collided
CFIT	mountain, mountains, mountainside, mount, mountainous, terrain, trees, struck, hillside, hill, controlled flight, ocean, lake, river, sea, wooded, jungle, mt, ravine, ridge, peak, volcano, swamp, channel, foothills
Pilot Error	pilot error, tailspin, tail spin, lost control, nose dive, nosedive, overran, overloaded, uncontrolled, unable, premature, procedure, spin, stall, improperly, inadequate, alcohol, unqualified, sharp turn, turned over, error, improper, procedures, stalled, maintain, failed to, neglected
Undetermined	cause unknown, cause undetermined, undetermined, reasons unknown, disappeared, never found, cause not determined, missing, wreckage, disintegrated
Unknown	No matching keywords found

Table 2: Contributing Factors and Associated Keywords

To verify this method, a representative subset was taken from the total population to be manually labeled. The size of this subset was determined using Cochran's Sample Size Formula for Finite Populations for a 95% Confidence Interval [6]. For a total dataset of 5268 accidents, a

representative subset is 358 accidents. Refer to Appendix A: Cochran's Sample Size Formula for Finite Populations for the details. After manually labeling the subset, a confusion matrix was generated to assess the quality of the classification system. The table detailing the results can be seen in Table 3.

Category	TP	TN	FP	FN	Precision	Recall	F1
sabotage	2	353	2	1	0.500	0.667	0.571
shot down	9	345	1	3	0.900	0.750	0.818
fire	44	305	9	0	0.830	1.000	0.907
fuel	8	350	0	0	1.000	1.000	1.000
weather	101	249	3	5	0.971	0.953	0.962
collision	11	341	4	2	0.733	0.846	0.786
cfit	133	204	14	7	0.905	0.950	0.927
mechanical	73	244	28	13	0.723	0.849	0.781
pilot error	73	249	8	28	0.901	0.723	0.802
undetermined	6	351	0	1	1.000	0.857	0.923
unknown	50	288	19	1	0.725	0.980	0.833

Table 3: Keyword Classification Confusion Matrix

The accuracy of the labeler was assumed sufficient for the purposes of this project. Improvements could be made utilizing possible natural language processing (NLP) approaches, or manual review of each datapoint. Categories that suffered such as sabotage is likely because they are present so little within the dataset that making adjustments did not improve performance of sabotage without decreasing the identification effectiveness of the other contributing factors.

Labeling Crash Location Data

The dataset also encodes the location of the crash of each fatal accident. This data is unstructured and is not classified together into groups that can be used to draw correlations between locations and accident severity, because the town specific data is too granular. Therefore, the crash locations were divided into subregions of the world using the 22 Subregions defined in the M49 Standard on Area Classifications for Statistical Use defined by the United Nations [7].

Using the same methodology as labeling the contributing factors, the location datapoints were searching for keywords that match locations that fit into the rules for each subregion. These take the forms of countries, or states in the United States. The keyword rules are available for reference in Appendix B: Subregion Keywords.

Finding the Casual Structure

To find the underlying casual structure that contributes to Aircraft Crashes and associated Fatalities, there is a combination of observed and unobserved factors. The observed factors will come from the Aircraft Fatalities dataset. The data that can be observed, although not perfectly, can be seen in Table 1. In addition, additional data can be derived from the information available

in Table 1 as explained previously. For example, the data details the location of each accident, and most flights details the origin and destination. This location data can be grouped into specific flight routes, or continents of operation. This can answer questions like, “How strong is the correlation between flight accidents in areas associated with active conflict, such as the middle east, and aircraft fatalities with a contributing factor of being shot down?” For the initial exploration into finding a causal link between the dataset values and the aircraft fatalities, the first approach will construct a naïve model between each of our casual factors, and the normalized rate of fatalities.

$$\hat{f} = \frac{\text{Aircraft Fatalities} + \text{Ground Fatalities}}{\text{Souls Aboard}}$$

Equation 1: Normalized Fatalities Equation

For the purposes of using the FCI with Chi-Squared Algorithm, all metric need to be discretized into either true or false. For the normalized fatalities, greater than 0.5 will yield a “high fatality = true” encoding.

Solving for the PAG

To determine the causal link between the recorded contributing factors, the library known as casual-learn [2], a popular Python Library with implementation of algorithms to solve for the adjacency matrix from data. The casual-learn implementation of the Fast Causal Inference Algorithm (FCI) [2] was utilized to create PAGs, partially ancestral graphs, from observational data.

The FCI Algorithm requires the following assumptions:

1. Casual Markov Condition – That every variable in independent of it’s non decedents.
2. Faithfulness – That independence in data arise from real causal structures.
3. No Selection Bias – That the data sample is not commonly conditioned on the variable on interest.

This is notable because other common algorithms such as the PC algorithm also requires causal sufficiency. It can inter latent confounder within the data. This is useful for our use case if there were confounding contributing factors present that may not have been noted either in the summary of each accident or accounted for in the labeling process.

Causal Discovery

Contextual Knowledge Derived Constraints

The causal discovery process involved the evaluation of each iteration output against contextual knowledge to identify structural artifacts: edges or orientations that are discovered by the algorithm to be present in the data but logically indefensible. These artifacts likely arrive out of a violation of the sampling bias or a violation of faithfulness. Each constraint imposed on the

model was motivated by a specific artifact observed in a prior iteration and justified by reasoning external to the dataset. This section describes these constraints and their associated justifications.

First, accidents in which exactly one person was aboard, and one fatality was recorded were excluded from the analysis. These records have a normalized fatality rate of exactly 1.0 by construction, but this reflects a selection artifact of the dataset rather than meaningful information about accident severity. Had these accidents resulted in zero fatalities, they would not appear in the dataset at all, since the dataset contains only fatal accidents. Their inclusion would bias the severity distribution upward without contributing meaningful variation, so they are treated as uninformative and removed.

Second, two variables were constrained to act as source-only nodes based on their exogenous nature. In the unconstrained model, weather appeared as an effect of other contributing factors, for instance, the graph implied that controlled flight into terrain causes adverse weather. Weather is an exogenous environmental condition determined independently of the accident mechanism and was therefore constrained to have only outgoing edges. Similarly, in early iterations the “is_military” variable appeared as an effect of contributing factors such as mechanical failure, implying that accident characteristics cause an aircraft to be operated by a military organization. Whether an aircraft is operated by a military organization is determined prior to and independently of the accident, so military status was likewise constrained to have only outgoing edges.

Third, high lethality was constrained as a sink node, receiving edges but producing none. The unconstrained graph contained edges in which high lethality appeared to cause contributing factors, such as high lethality causing mechanical failure. The severity of an accident is a downstream consequence of the contributing factors and the circumstances of the crash; it cannot retroactively cause the factors that preceded it. Constraining high lethality to receive only edges ensures that it is modeled exclusively as an outcome variable.

Fourth, the undetermined category was excluded from the causal model entirely. In the constrained iterations, undetermined participated in causal edges alongside substantive contributing factors, but it does not represent an actual mechanism, it merely indicates that investigators were unable to conclusively identify the cause. Retaining it would introduce a category error, as the model would be estimating the causal effect of failing to determine a cause on accident severity, which is not a meaningful causal question. Accidents classified as undetermined remain in the dataset, but their contributing factor label does not participate in the causal structure discovery.

Next, the preliminary iterations used the PC algorithm, which assumes causal sufficiency. This assumption is difficult to defend for aviation accident data, where plausible unmeasured confounders include pilot experience, aircraft age, maintenance quality, regulatory environment, operational risk tolerance, and terrain characteristics. The final model was therefore estimated using the Fast Causal Inference (FCI) algorithm with the chi-square conditional independence test. FCI relaxes the causal sufficiency assumption and permits the representation of latent confounders through circle-endpoint edge marks in the resulting PAG, allowing the model to

express uncertainty about whether observed associations reflect direct causation or confounding by unmeasured unobserved variables.

Then, upon introducing UN M49 subregion as a geographic variable, two additional constraints were required. The algorithm initially produced an edge from weather to subregion, implying that weather conditions determine the geographic region in which an accident occurs. The causal direction is the reverse; regional geography and climate patterns influence the weather conditions encountered during flight, so a background constraint was encoded requiring the direction to flow from subregion to weather. In the subsequent iteration, the algorithm produced an edge from CFIT to subregion, again implying that the contributing factor determines where the accident occurs. Subregion is exogenous to the accident mechanism; it is determined by the flight's route and operational context. The expected causal direction is that subregion influences CFIT through the unobserved mediator of local terrain characteristics, such as mountainous regions that produce higher rates of controlled flight into terrain. A constraint was encoded accordingly to enforce that edge direction. The edge from military status to subregion was retained without correction, as military mission requirements plausibly determine the operating theater in which the aircraft flies.

The pilot error node was constrained to have a directed edge into CFIT. In the graph produced after the subregion corrections, pilot error appeared as a pure sink node with no outgoing edges, which contradicts the definition of CFIT, a scenario in which a qualified crew flies an airworthy aircraft into terrain, inherently involving errors in situational awareness, navigation, or altitude management. The absence of this edge is an artifact of selection bias: because the dataset contains only fatal accidents, both pilot error and CFIT occur at elevated base rates relative to the general population of flights, compressing the conditional dependence that would be observable in an unbiased sample. Encoding this relationship ensures that the graph reflects the well-established definitional link between pilot error and controlled flight into terrain.

The final adjustment necessary to complete the graph was to fix the direction for sabotage. FCI found the edge direction between Sabotage and fire to be ambiguous. The causal relationship was assumed to be that sabotage causes fire to be reported in the summary.

Causal Graph – Directed Acyclic Graph

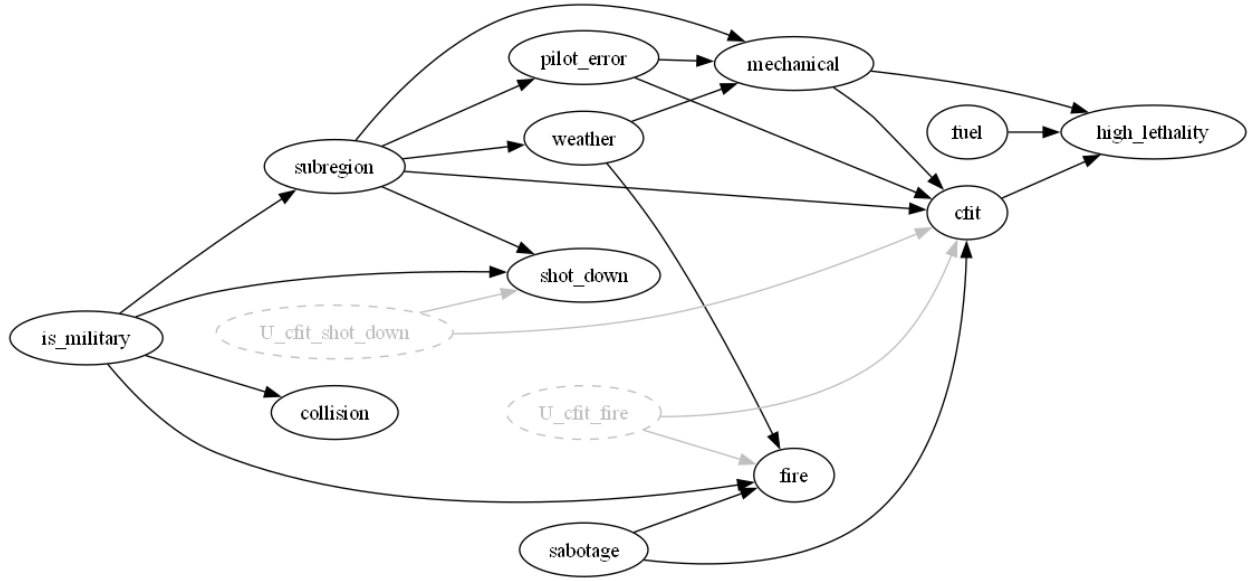


Figure 1: DAG derived from PAG using the FCI Algorithm with Enforced Constraints

Using the FCI Algorithm implementation in DoWhy, Figure 1 shows the resulting DAG, which is assumed to represent causal relationships. The next step was to encode this graph into DoWhy, where adjustment sets could be identified for each treatment variable.

Adjustment Sets for each Treatment Variable

Solving the Causal Impact from Estimated PAG

The next step of analysis is to quantify the causal impact of each factor on lethality using linear regression. The PAG, which has most edges to be directed, can be used to determine which variables should be controlled to isolate the causal path between each node. First, all open paths between the variables need to be identified and then closed. Table 1 Table 4 shows all the relationships that will be studied for causal effect with linear regression and the discovered adjustment sets. Initial focus is on the principal sink node, which is accident lethality. The backdoor adjustment sets are conservative as a byproduct of DoWhy's backdoor adjustment identification tool, by which it conditions on all non-decedents to block backdoor paths, it does not output the minimal adjustment set.

Treatment	Approach	Adjustment Set	Outcome
is_military	No Backdoor Paths	[]	$P(\hat{f} > 0.5)$
collision	No Path	[]	$P(\hat{f} > 0.5)$

cfit	Backdoor Adjustment	['fire', 'pilot_error', 'sabotage', 'collision', 'weather', 'is_military', 'subregion']	$P(\hat{f} > 0.5)$
fire	No Path	[]	$P(\hat{f} > 0.5)$
mechanical	Backdoor Adjustment	['fire', 'pilot_error', 'sabotage', 'collision', 'weather', 'is_military', 'subregion', 'shot_down', 'fuel']	$P(\hat{f} > 0.5)$
pilot_error	Backdoor Adjustment	['fire', 'sabotage', 'collision', 'weather', 'is_military', 'subregion', 'shot_down', 'fuel']	$P(\hat{f} > 0.5)$
weather	Backdoor Adjustment	['pilot_error', 'sabotage', 'collision', 'is_military', 'subregion', 'shot_down', 'fuel']	$P(\hat{f} > 0.5)$
shot_down	No Path	[]	$P(\hat{f} > 0.5)$
sabotage	No Backdoor Path	[]	$P(\hat{f} > 0.5)$
Fuel	No Backdoor Path	[]	$P(\hat{f} > 0.5)$
subregion	No Backdoor Path	[]	$P(\hat{f} > 0.5)$
is_military	No Backdoor Path	[]	$P(\hat{f} > 0.5)$

Table 4: Relationships to Quantify and Adjustment Sets

The OLS Model to be Fit

$$\begin{aligned}
& \text{high_lethality} \\
&= \beta_0 + \beta_{mil} \cdot is_military + \beta_{sub} \cdot subregion + \beta_{cfit} \cdot cfit + \beta_{col} \\
&\cdot collision + \beta_{fire} \cdot fire + \beta_{fuel} \cdot fuel + \beta_{mec} \cdot mechanical + \beta_{pe} \\
&\cdot pilot_error + \beta_{sab} \cdot sabotage + \beta_{sd} \cdot shot_down + \beta_{wea} \cdot weather + \varepsilon
\end{aligned}$$

Equation 2: OLS Model to be fit for causal discovery

$$\beta = (X^T X)^{-1} X^T y$$

Equation 3: Solution for OLS

This model is solved for to identify the causal impact of each factor in making an already lethal accident “highly lethal”. For example, if β_{mil} is found to be 0.20, that would mean in an accident where that factor was present ($is_military = true$), the accident would have a 20pp greater chance of being classified as “highly lethal”. To clarify, 20pp is not equal to 20% greater chance. If the chances, all things being equal were previously 10%, and the coefficient is 20pp, the result would be a 30% chance. That is a 200% increase.

The Results

Data Preparation and Classification

Treatment	OLS Coeff	p-value	P<0.05
subregion	0.0015	0.2091	FALSE
is_military	0.1624	<0.0001	TRUE
cfit	0.0664	<0.0001	TRUE
collision	n/a	n/a	n/a
fire	n/a	n/a	n/a
fuel	-0.2253	<0.0001	TRUE
mechanical	-0.0779	<0.0001	TRUE
pilot_error	-0.0239	0.0642	FALSE
sabotage	-0.1593	0.0001	TRUE
shot_down	n/a	n/a	n/a
weather	0.0293	0.0199	TRUE

Table 5: OLS Coefficients and p-values

Here is the result of OLS on the model described above. Of the 11 parameters regressed, 3 did not have a causal path that was discovered in the derived DAG, and two were not statically significant ($p\text{-value} < 0.05$). Of the remaining, we have `is_military`, `cfit`, `fuel`, `mechanical`, `sabotage`, and `weather`. The negative Coefficients mean that factor being present is associated with less severe accidents, and those with positive, more severe accidents. The conclusion sections will discuss possible interpretations of these results. However, the classification system based on the use of keywords succeeded in grouping summaries of aviation accidents according to eleven categories of contributing factors, thus facilitating the analysis of unstructured text data. The best performance was achieved by `weather`, `CFIT`, and `fire` because of the high values of their F1 score. In these cases, there is the same clear wording (for example, storm, terrain, explosion) that helps to achieve a high level of classification quality.

In contrast, `sabotage` and `unknown` proved to have poorer results in terms of classification quality because of the vagueness of descriptions and the ambiguity of information in the dataset. Often the summaries do not include relevant details, and thus the classification process may be quite difficult. Nevertheless, despite its shortcomings, this classification system provides an adequate base for further causal analysis since the most important contributors were identified with high precision.

Causal Structure Modeling

In the initial causation networks constructed by the algorithm, there were some unrealistic dependencies; for instance, `weather` being modeled as the dependent variable of other variables. This can be attributed to the limitation of such algorithms on observational data in the absence of constraints since causal discovery algorithms operate purely based on independence

rather than domain knowledge. This problem was addressed using domain knowledge-based constraints. In particular, weather conditions and military service were modeled as exogenous nodes, whereas accident lethality was restricted to be a sink node. Also, the category “undetermined” was excluded from consideration due to its ambiguous nature. As a result, a network in the form of DAG was obtained which models the realistic dynamics in the aviation industry. From the experiment, even though causal discovery algorithms are robust, they require substantial subject matter expertise for obtaining meaningful outputs.

Key Causal Findings

Some of the key correlations observed through the developed causal model include: Weather conditions were seen to increase the chances of human error among pilots. The reasons included the difficulties in handling the flights during such times due to increased turbulence and poor visibility. The correlation between pilot error and accidents related to collisions with objects and terrain was also observed. Both types of accidents often result from loss of situational awareness and poor judgment. Controlled Flight into Terrain became a critical contributing factor to high lethality. Such accidents occur during impact with terrain of controlled flights. This means that there is very little chance for survival after such accidents happen. Mechanical failures became another contributing factor to high lethality. Such failures cause the loss of aircraft control and cannot be recovered from easily. The association between fuel-related failures and lethality appeared to be less strong and inconsistent compared to the others. The reason behind this could be the fact that, in case of fuel exhaustion or starvation, the pilot can still make an emergency landing. Unlike other types of failures, they do not lead to an immediate accident or explosion. Another interesting finding made by the model was the association between military airplanes and risky conditions. It is unlikely that this is a purely causal connection because there might be some unknown variables involved in the process that influence this relation and are not considered in the dataset used for analysis.

Unexpected Relationships and Limitations

There were some other relations found in the causal graph that were quite surprising or hard to understand. First, there was supposed to be a connection between sabotage and weather conditions, but this relation would hardly qualify as a causal one and could result from biases in the data collection process or the presence of confounding variables. Also, another surprising aspect of this analysis is that there is only a slight relation between collisions and fatality. It might be due to the fact that there are different kinds of collisions, varying greatly in intensity and results. It means that we have some problems with the selected data. First, it includes only fatal crashes, creating a selection bias, while comparisons of lethal and non-lethal accidents would help to find additional patterns in the data. Also, there is no information on such factors as aircraft design and mission conditions.

Causal Impact on Lethality

Estimating causal effects based on regression shows that there are many causes of accidents that make them lethal, but the relationship is strongly influenced by interactions between them. Weather, for example, does not always lead to a fatal crash in aviation, but the interaction of weather-related pilot error greatly increases chances of such an outcome. However, there is evidence that the relationship between factors like CFIT or mechanical malfunction is more direct, thus leading to fatality. The findings once again confirm that aviation accidents are indeed multifactorial in nature, with outcomes resulting from complicated interactions between different factors. In summary, using the methodology of causal structural analysis combined with estimation of effects provides deeper insights into aviation accidents that could not have been reached otherwise.

The Conclusion

This study applied a causal inference framework, combining the FCI algorithm, PAG-based causal discovery, and OLS regression with backdoor adjustments -- to a dataset of 5,268 fatal aviation accidents spanning 1908 to 2009. The approach moves beyond descriptive correlation by leveraging Pearl's do-calculus framework to isolate the causal contribution of each identified factor to accident lethality.

Of the eleven contributing factors examined, six produced statistically significant causal estimates. Military operation status and Controlled Flight into Terrain (CFIT) were the strongest positive drivers of high-lethality outcomes, increasing the probability by 16.2 and 6.6 percentage points, respectively. Adverse weather contributed a smaller but significant 2.9 percentage point increase. These results are consistent with domain knowledge: military missions, terrain impacts, and poor visibility all reduce the window for corrective action and survivable crash conditions.

In contrast, fuel-related incidents and mechanical failure were associated with significantly lower lethality, by 22.5 and 7.8 percentage points, respectively. The most likely explanation is that these failure modes are frequently progressive rather than instantaneous, affording flight crews time to recognize the problem, declare an emergency, and execute a controlled forced landing. This finding has actionable implications: investments in early-warning systems for fuel management and mechanical monitoring may not reduce accident frequency but could meaningfully improve survival rates when accidents do occur.

The negative coefficient for sabotage warrants caution in interpretation. The sabotage category had the weakest classifier performance ($F1 = 0.571$), and its low prevalence in the dataset likely results in unstable estimates. Future work should refine this category using NLP-based classification to improve labeling fidelity.

Several limitations of this study should be acknowledged. The dataset is restricted to fatal accidents, which introduces selection bias and limits the ability to estimate absolute crash risk;

only relative severity can be assessed. Additionally, latent confounders such as pilot experience, aircraft age, and regulatory era are unobserved and not fully accountable even under FCI. The use of background constraints to correct logically indefensible edges, while principled, introduces researcher degrees of freedom into the causal discovery process.

Nonetheless, the causal framework applied here demonstrates clear analytical advantages over traditional correlation-based aviation safety analysis. By identifying the structural relationships between contributing factors rather than treating them as independent predictors, the model surfaces causal drivers rather than coincidental associations. The findings suggest that future aviation safety research and regulatory policy would benefit substantially from the wider adoption of causal discovery methods alongside conventional statistical approaches.

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Appendix A: Cochran's Sample Size Formula for Finite Populations

Let Z represent the dataset.

$$N = |A| = 5268$$

$$a \in A$$

$$n = |a|$$

$$p = 0.5, q = 0.5$$

$$e = 0.05$$

$$Z_{score}(95\% CI) = 1.96$$

$$n_0 = \frac{Z^2 pq}{e^2} = \frac{(1.96)^2 (0.5)(0.5)}{(0.05)^2} = 385$$

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}} = \frac{385}{1 + \frac{385 - 1}{5268}} = 358$$

Appendix B: Subregion Keywords

Sub-Region	Keywords
Australia and New Zealand	australia, new zealand, tasmania
Caribbean	antigua, bahamas, barbados, british virgin islands, caribbean, cuba, dominican republic, dominica, dominican republic, french west indies, great inagua, grenadines islands, guadeloupe, haiti, hati, jamacia, jamaica, leeward islands, martinique, netherlands antilles, puerto rico, saint lucia, saint lucia island, sainte lucia island, trinidad, turks & caicos islands, turks and caicos islands, u.s. virgin islands, us virgin islands, virgin islands, west indies
Central America	belize, costa rica, el salvador, guatemala, honduras, mexico, nicaragua, panama
Central Asia	kazakhstan, kirghizia, kyrgyzstan, tajikistan, turkmenistan, uzbekistan
Eastern Africa	central mozambique, comoro islands, comoros, comoros islands, djibouti, djibouti, eritrea, ethiopia, french somaliland, kenya, madagascar, malagasy republic, malawi, mocambique, mozambique, reunion, rhodesia, rwanda, somalia, south sudan, tanganyika, tanzania, uganda, zambia, zimbabwe
Eastern Asia	china, formosa strait, hong kong, japan, mongolia, north korea, okinawa, south korea, taiwan
Eastern Europe	bugaria, bulgaria, bulgeria, chechnya, czech republic, czechoslovakia, east germany, hunary, hungary, moldova, poland, romania, romania, russia, russian, slovakia, soviet union, ukraine, ussr
Melanesia	fiji, new guinea, papua new guinea, solomon islands, vanuatu
Micronesia	guam, mariana islands, marshall islands, wake island
Middle Africa	angola, belgian congo, belgium congo, cameroon, cameroons, central african republic, chad, congo, democratirepublicogo, democratirepublicongo, democratirepublicof congo, democtratirepublicongo, dr congo, equatorial guinea, french cameroons, french equatorial africa, french equitorial africa, gabon, zaire
Northern Africa	algeria, algiers, egypt, libya, morocco, morocco, morroco, morroco, sudan, tunisia, uar
Northern America	airzona, ak, al, alabama, alaksa, alakska, alaska, alberta, ar, arazona, arizona, arkansas, az, bermuda, british columbia, british columbia canada, ca, cailifornia, california, california, canada, co, coloado, colorado, connecticut, ct, d.c., dc, de, delaware, deleware, fl, florida, ga, georgia, greenland, hawaii, hi, ia, id, idaho, il, ilinois, illinois, in, indiana, iowa, kansas, kentucky, ks, ky, la, labrador, louisana, louisiana, ma, maine, manitoba, maryland, massachusett, massachusetts, massachutes, md, me, mi, michigan, minnesota, minnisota, mississippi, missouri, mn, mo, montana, ms, mt, nc, nd, ne,

	nebraska, nevada, new brunswick, new hampshire, new jersey, new mexico, new york, newfoundland, nh, nj, nm, north carolina, north dakota, nova scotia, nv, ny, oh, ohio, ok, oklahoma, oklohoma, ontario, or, oregon, pa, pennsylvania, quebec, rhode island, ri, saskatchewan, sc, sd, south carolina, south dakota, south dekota, tennessee, tennessee, texas, tn, tx, united states, usa, ut, utah, va, vermont, virginia, vt, wa, washington, washington, washington d.c., washington dc, west virginia, wi, wisconsin, wisconsin, wv, wy, wyoming, yukon, yukon territory
Northern Europe	denmark, england, estonia, finland, guernsey, iceland, ireland, islay island, isle of man, jersey, latvia, lithuania, northern ireland, norway, scotland, sweden, uk, united kingdom, wales
Polynesia	american samoa, cook islands, french polynesia, samoa, tahiti, u.s. samoa, west samoa, western samoa
South America	aregntina, argentina, boliva, bolivia, brazil, chile, chili, colombia, dutch guyana, ecuador, french guiana, guyana, paraguay, peru, surinam, suriname, uruguay, venezuela
South-eastern Asia	borneo, brunei, burma, cambodia, east timor, french indo-china, indonesia, inodnesia, khmer republic, laos, malaya, malaysia, manmar, myanmar, netherlands indies, philipines, philippines, philipines, sarawak, singapore, south vietnam, thailand, thiland, timor, vietnam
Southern Africa	botswana, lesotho, namibia, south africa
Southern Asia	afghanistan, baangladesh, bangladesh, bhutan, india, napal, nepal, pakistan, sri lanka, west pakistan
Southern Europe	albania, azores, bosnia, bosnia herzegovina, bosnia-herzegovina, canary islands, croatia, greece, italy, kosovo, macedonia, malta, portugal, serbia, slovenia, spain, yugosalvia, yugoslavia
Western Africa	benin, burkina faso, cape verde, cape verde islands, french west africa, gambia, ghana, guinea, ivory coast, liberia, mali, mauretania, mauritania, niger, nigeria, senegal, sierra leone, sierra leone, togo, upper volta
Western Asia	armenia, azerbaijan, bahrain, cyprus, iran, iraq, israel, jordan, kuwait, lebanon, oman, qatar, republic of georgia, republics of georgia, republiof georgia, saudi arabia, south yemen, syria, turkey, uae, united arab emirates, yemen
Western Europe	austria, belgium, france, germany, luxembourg, netherlands, switzerland, the netherlands, west germany

Appendix C: Causal Paths

